

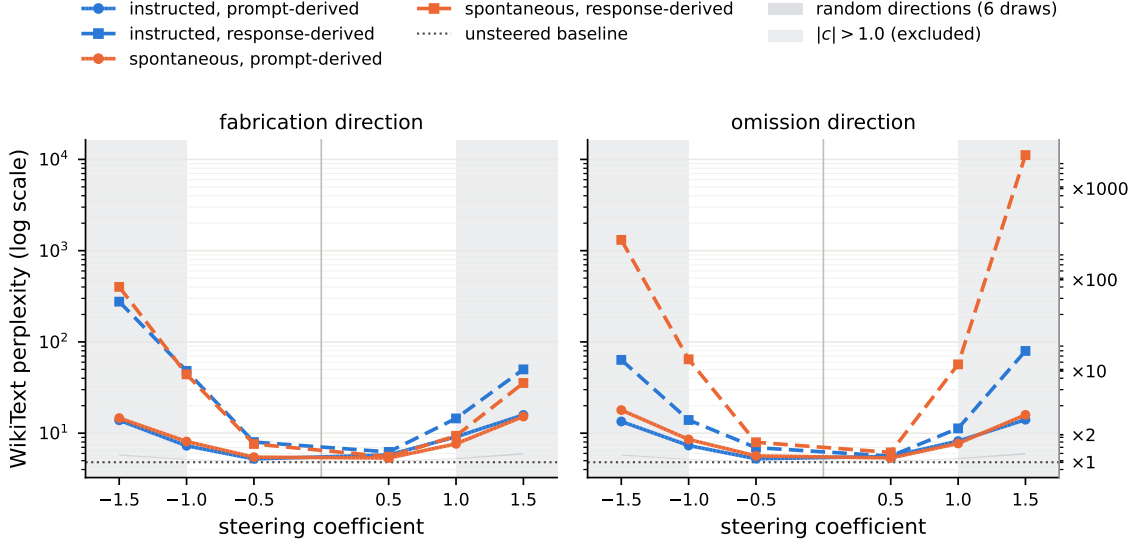


Figure 8: Perplexity scores on WikiText-2 under steering at layers 19-32 for all tokens. The y axis is logarithmic due to the dramatic scores some direction steering produced. The right hand axis gives multiples of the unsteered baseline.

Appendix F. Reproducibility

The repository for this paper can be found at <https://github.com/JosiahL98/Spont-I-nstructed-Deception>. The scaffolding for our repository builds on the codebase of Yang et al. (2024).

Experiments were run through NDIF, the OpenAI API, and OpenRouter API. The model studied was Llama-3.1-70B-Instruct, 80 layers, d model 8192. Residual stream activations were extracted through NNsight and NDIF. Greedy decoding was used throughout in generation by the model.

Gpt-5-mini was used via OpenRouter to generate the spontaneous and instructed deception prompts. Gpt-4.1-mini-2025-04-14 was used via the OpenAI API to generate the ratings of deception and coherence for the Model’s responses to the prompts.

Tables and figures in this paper were generated by scripts available in the repository. The mapping from data to tables and figures is listed in the paper runs manifest file in the repo.

Random seeds 43 and 44 were used in addition to 42 at layer 35 in probing experiments. Random directions were computed with seeds 0, 1, and 2. All other code using random seeds used seed 42.